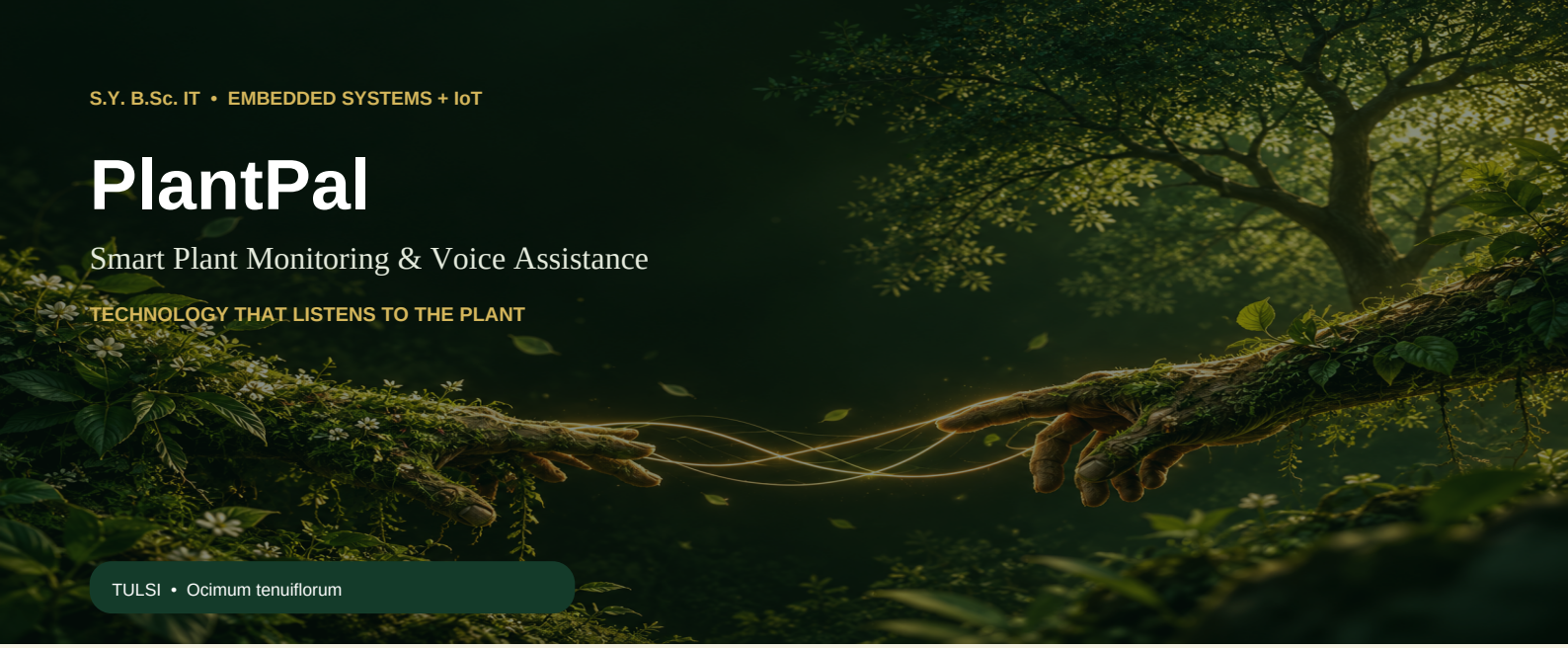


S.Y. B.Sc. IT • EMBEDDED SYSTEMS + IoT

PlantPal

Smart Plant Monitoring & Voice Assistance

TECHNOLOGY THAT LISTENS TO THE PLANT

TULSI • Ocimum tenuiflorum

THE PROBLEM

Plant care often relies on visual checks. Changes in soil moisture, light and surrounding conditions can go unnoticed.

PlantPal makes sensor readings easier to understand.

OUR APPROACH

A Tulsi plant is monitored by an ESP32 and low-cost sensors. The device interprets readings locally and shares selected data with a web dashboard.

Local feedback is designed to work without the cloud.

HOW IT WORKS

01 · SENSE — soil moisture, light, temperature, humidity and touch.

02 · PROCESS — ESP32 validates readings and classifies project-defined states.

03 · RESPOND — OLED, DFPlayer Mini voice prompts, speaker and buzzer.

04 · CONNECT — Wi-Fi → Flask REST API → SQLite → browser dashboard.

SYSTEM AT A GLANCE

Controller: ESP32-WROOM-32

Sensors: Capacitive soil · BH1750 light · DHT11 climate · TTP223 touch

Outputs: OLED · 73 stored voice tracks · buzzer

Cloud: HTTPS/REST · Flask · SQLite · web dashboard

Watering: human-controlled; no pump.

SYSTEM FLOW

SENSORS Read inputs › **ESP32** Classify › **LOCAL** OLED + voice › **CLOUD** Dashboard

KEY BENEFITS

- Multi-sensor readings in one device.
- Local screen, touch and event-based voice.
- Cloud monitoring and remote commands.

LIMITATIONS

- Soil percentage needs calibration in final soil.
- Thresholds are project-defined indicators.
- Human decides when to water.

PROJECT TEAM

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